

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

B.Tech Degree S7 (R, S) / S7 (PT) (R,S) Examination November 2024 (2019 Scheme)

**Course Code: EET401****Course Name: ADVANCED CONTROL SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- 1 The differential equation model of a system is given below. Obtain the state space model.  $4 \frac{d^3 c(t)}{dt^3} + 3 \frac{d^2 c(t)}{dt^2} + \frac{dc(t)}{dt} + 2c(t) = 5r(t)$  (3)
- 2 Obtain the state space model of the system in diagonal canonical form whose transfer function is given by  $\frac{C(s)}{R(s)} = \frac{2(s+3)}{(s+1)(s+2)}$  (3)
- 3 The state space model of a system is given by  $\dot{X} = \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix} X + \begin{bmatrix} 1 \\ 1 \end{bmatrix} U$  and  $Y = [1 \ 0]X$ . Determine the transfer function of the system. (3)
- 4 Differentiate zero input response and zero state response of a system. (3)
- 5 Briefly explain full order state observer and reduced order state observer. (3)
- 6 Illustrate the necessary and sufficient condition for state feedback controller design. (3)
- 7 What is limit cycle? What are the means to find the existence of limit cycles in describing function analysis. (3)
- 8 List any two common nonlinearities present in the system and draw the respective input output response of each nonlinearity. (3)
- 9 With a neat diagram, explain phase plane and phase trajectory. (3)
- 10 How do we choose Lyapunov functions for nonlinear systems. (3)

**PART B***Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) Obtain the state model of the electrical network shown in figure 1. Select current in resistor  $R_2$  as output. (8)

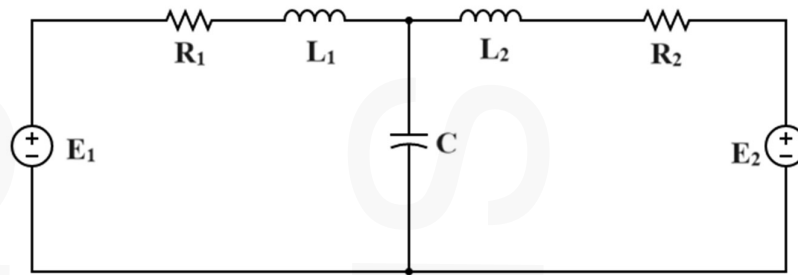


Figure 1

- b) For the system given below, derive the state space model in phase variable form. (6)

$$\frac{Y(s)}{U(s)} = \frac{2s^2 + 6s + 5}{s^3 + 4s^2 + 5s + 2}$$

OR

- 12 a) For the network shown in figure 2, obtain the state equation and output equation. (6)

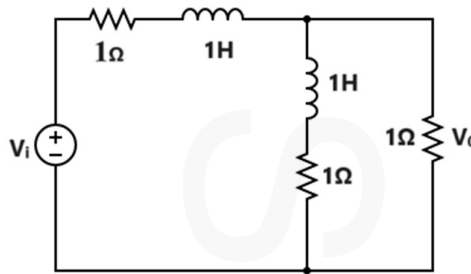


Figure 2

- b) Obtain the state model in canonical form for the system described by the differential equation,  $\ddot{y} + 6\dot{y} + 11y = \ddot{u} + 8\dot{u} + 17u + 8u$  (8)

Module II

- 13 a) A linear time invariant system is characterized by the state equation, (10)

$$\dot{X} = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} U. \text{ Where } U \text{ is the unit step input. Obtain the solution of the state equation with following initial condition } X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

- b) Write the properties of state transition matrix of discrete time system. (4)

OR

- 14 a) Using Cayley Hamilton method, determine the state transition matrix for (9)

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$$

- b) Obtain the solution of nonhomogeneous state equation  $\dot{X}(t) = AX(t) + BU(t)$  (5) using Laplace transform method.

## Module III

- 15 a) Evaluate the observability of the system with  $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix}$ ,  $B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$  and  $C = [3 \ 4 \ 1]$  using Gilberts test. (6)

- b) Consider the system described by the state model  $\dot{X} = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} U$  and  $Y = [1 \ 0]X$ . Design a full order state observer such that the desired eigen values for the observer are -5,-5. (8)

OR

- 16 a) For the system characterized by the state equation given below, design a state feedback controller to place the closed loop poles of the system at  $s = -2 \pm j4$  and  $s = -10$ . (10)

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix} X + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} U$$

- b) Explain how we can check the controllability of a system using PBH test (4)

## Module IV

- 17 a) What is meant by Describing function? Derive the describing function of a saturation nonlinearity. (8)

- b) Explain the assumptions made referred to describing function analysis. (6)

OR

- 18 a) Consider the nonlinear system of Figure 3, with a saturating amplifier having gain K in its linear region. Determine the largest value of gain K for the system to stay stable. What would be the frequency, amplitude, and nature of the limit cycle for a gain K=3. (14)

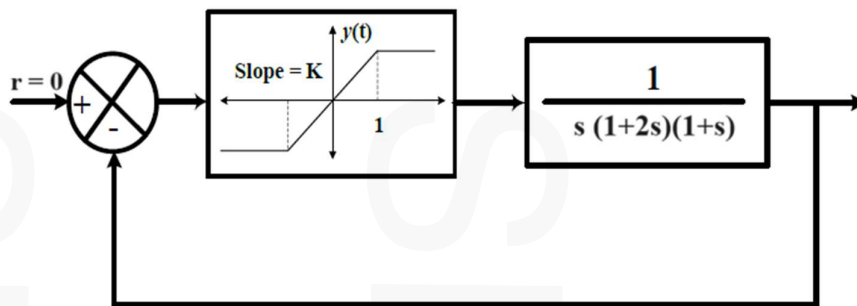


Figure 3

## Module V

- 19 a) Construct the phase trajectory of the second order system given below using isocline method.  $\frac{d^2x}{dt^2} + \frac{dx}{dt} + x = 0$ . (7)

The initial condition of the system is given as  $X(0) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

- b) Compute the Lyapunov function,  $V(X)$  for which the system given below is asymptotically stable.  $\dot{X} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} X$  (7)

OR

- 20 a) Consider the following nonlinear system,  $\dot{x}_1 = x_2$  and  $\dot{x}_2 = -\sin(x_1) - x_2$ . (9)

Find all singular points of the system, classify them, and sketch the phase portrait in the neighbourhood of singular points.

- b) Investigate the stability of the following non-linear system using Lyapunov direct method.  $\dot{x}_1 = x_2$  and  $\dot{x}_2 = -x_1^2 x_2$  (5)

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